

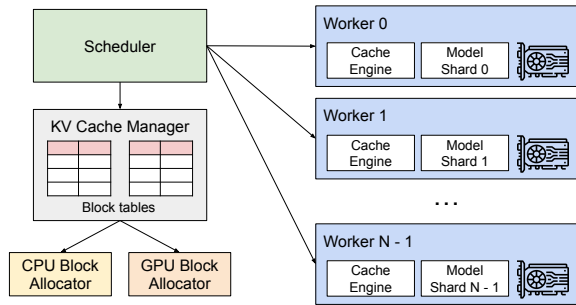


Figure 4. vLLM system overview.

Although compaction [54] has been proposed as a potential solution to fragmentation, performing compaction in a performance-sensitive LLM serving system is impractical due to the massive KV cache. Even with compaction, the re-allocated chunk space for each request prevents memory sharing specific to decoding algorithms in existing memory management systems.

Method

In this work, we develop a new attention algorithm, *PagedAttention*, and build an LLM serving engine, *vLLM*, to tackle

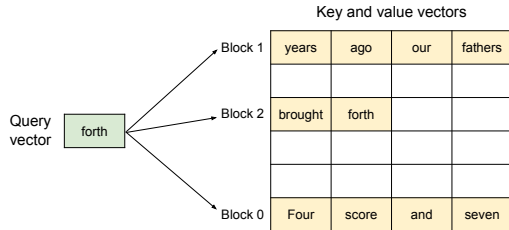


Figure 5. Illustration of the PagedAttention algorithm where the attention key and values vectors are stored across non-contiguous blocks in the memory.

block size (B). Denote the key block $K_j = (k_{(j-1)B+1}, \dots, k_{jB})$ and value block $V_j = (v_{(j-1)B+1}, \dots, v_{jB})$. The attention computation in Eq. 4 can be transformed into the following block-wise computation:

$$A_{ij} = \frac{\exp(q_i^\top K_j / \sqrt{d})}{\sum_{t=1}^{\lceil i/B \rceil} \exp(q_i^\top K_t \mathbf{1} / \sqrt{d})}, \quad o_i = \sum_{j=1}^{\lceil i/B \rceil} V_j A_{ij}^\top, \quad (4)$$

where $A_{ij} = (a_{i,(j-1)B+1}, \dots, a_{i,jB})$ is the row vector of attention score on j -th KV block.

During the attention computation, the PagedAttention kernel identifies and fetches different KV blocks separately